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Management of Abnormalities of the Genitalia in Girls

Martin Kaefer

Keywords

clitoris; complete androgen insensitivity syndrome; congenital adrenal hyperplasia; disorders of sexual differentiation; female circumcision; Gartner duct cyst; Herlyn-Werner-Wunderlich syndrome; imperforate hymen; infibulation; labial adhesions; Male genital embryology; MRKH syndrome; neovagina; prolapsed ureterocele; rhabdomyosarcoma; transverse vaginal septum; urethral prolapse; vagina; vaginal atresia; vaginal genesis

Questions

1. Which of the following statements is TRUE regarding Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome?
    a. Patients present most commonly with infertility.
    b. It is a homogeneous disorder entailing congenital absence of the uterus and vagina.
    c. It is associated with a spectrum of ovarian abnormalities.
    d. It has associated upper urinary tract anomalies, primarily with the atypical disorder.
    e. It is associated with persistent Wolffian duct structures.
2. What is the crucial period in embryogenesis for the formation of the terminal bowel, kidney, paramesonephric ductal system, and

lumbosacral spine?

a. 4 to 6 weeks
b. 8 to 10 weeks
c. 10 to 14 weeks
d. 14 to 18 weeks
e. After 18 weeks

3. Which of the following is FALSE regarding vaginal agenesis (Müllerian aplasia)?

a. It occurs in approximately 1 in 5000 live female births.
b. Serum follicle-stimulating hormone and luteinizing hormone levels can be expected to be abnormally high.
c. Embryologically, it results from a failure of the sinovaginal bulbs to develop and form the vaginal plate.
d. It is associated with renal abnormalities.
e. It is associated with skeletal abnormalities.

4. Skeletal anomalies are found in what percentage of patients with Meyer-Rokitansky-Küster-Hauser syndrome?

a. 10% to 20%
b. 25% to 35%
c. 40% to 60%
d. 70% to 90%
e. 0% (they are not seen in MRKH)

5. What is the most common cause of primary amenorrhea?

a. Complete androgen insensitivity syndrome
b. Vaginal agenesis
c. Mixed gonadal dysgenesis
d. Imperforate hymen
e. Transverse vaginal septum

6. Which of the following statement is FALSE regarding the genitalia of women with Meyer-Rokitansky-Küster-Hauser syndrome?

a. In approximately 10% of patients, a normal but obstructed uterus or rudimentary uterus with functional endometrium is present.
b. Normal fallopian tubes are seen in approximately 35% of patients.
c. The ovaries are not functional in the majority of patients.

d. The hymenal ridge is usually present, along with a small vaginal pouch.
e. The labia majora are typically normal in appearance.

7. Uterus didelphys with unilateral imperforate vagina is most commonly present with which condition?
   a. Primary amenorrhea
   b. Cyclic abdominal pain associated with normal cyclic menstruation.
   c. Renal anomalies contralateral to the side of the obstruction
   d. Anomalies of the axial skeleton
   e. Constipation
8. Urethral prolapse is most commonly seen in young females of which ethnic background?
   a. African American
   b. White
   c. Asian
   d. Hispanic
   e. Native American
9. In most cases of labial adhesions, which of the following is true?
   a. They are believed to occur because of a relative state of hyperestrogenism.
   b. They should be treated with surgical lysis.
   c. They require no treatment.
   d. They occur secondary to sexual abuse.
   e. They have associated renal anomalies.
10. What is the mean age of a child with vaginal rhabdomyosarcoma?
   a. Younger than 2 years
   b. 2 to 4 years
   c. 4 to 8 years
   d. 8 to 12 years
   e. Older than 12 years
11. All of the following statements are true of anogenital condyloma acuminatum EXCEPT:
   a. human papillomavirus is the etiologic agent.
   b. sexual abuse is the only means by which an infant can contract the disease.

c. many cases in children resolve spontaneously.
d. pediatric immunization should dramatically reduce the incidence of this disorder.
e. carbon dioxide laser ablation is successful for anogenital lesions.

Answers

1. **d. It has associated upper urinary tract anomalies, primarily with the atypical disorder.** Urinary tract anomalies occur more commonly in patients with the atypical form of the disorder than in patients with the typical syndrome.
2. **a. 4 to 6 weeks.** Laboratory data with teratogens support the concept of a key event occurring between the fourth and fifth weeks of gestation that results in an error in the simultaneous development of the terminal bowel, kidney, bladder, paramesonephric ductal system, and lumbosacral spine.
3. **b. Serum follicle-stimulating hormone and luteinizing hormone levels can be expected to be abnormally high. Vaginal agenesis, which occurs in approximately 1 in 5000 live female births, is the congenital absence of the proximal portion of the vagina in an otherwise phenotypically (i.e., normal secondary sexual characteristics), chromosomally (i.e., 46,XX), and hormonally (i.e., normal luteinizing hormone and follicle-stimulating hormone levels) intact female.** It results from a failure of the sinovaginal bulbs to develop and form the vaginal plate. **Hauser brought further attention to the frequent association of renal and skeletal anomalies in these patients and stressed the differences between patients with these findings and those with complete androgen insensitivity syndrome (testicular feminization).**
4. **a. 10% to 20%.** Associated congenital abnormalities of the skeletal system have been described in 10% to 20% of cases.
5. **c. Mixed gonadal dysgenesis.** Meyer-Rokitansky-Küster-Hauser syndrome is in fact secondary only to gonadal dysgenesis as a

cause of primary amenorrhea.

6. **c. The ovaries are not functional in the majority of patients.** Although occasionally cystic, the ovaries are almost always present and functional.
7. **b. Cyclic abdominal pain associated with normal cyclic menstruation.** As with other obstructive disorders, the patient may present with cyclical or chronic abdominal pain. However, unlike other obstructive processes, duplication anomalies with unilateral obstruction are not associated with primary amenorrhea.
8. **a. African American.** This entity, which was first described by Solinger in 1732, occurs most often in prepubertal black girls and postmenopausal white women.
9. **c. They require no treatment.** Most children do not require treatment unless one of the aforementioned symptoms (urine pooling within the vagina, which may lead to postvoid dribbling; urinary tract infection; perineal irritation; physical findings of sexual abuse) occurs.
10. **a. Younger than 2 years.** The mean age of patients with primary vaginal tumors is younger than 2 years.
11. **b. Sexual abuse is the only means by which an infant can contract the disease.** Although a very high suspicion for sexual abuse is warranted, it should be kept in mind that perinatal transmission is also a possible mechanism. The ideal treatment depends on patient age and location and severity of the lesions.

Chapter Review

1. Remnants of the prostatic ducts give rise to the Skene glands; remnants of the Wolffian ducts give rise to the Gartner ducts.
2. The Bartholin glands are homologues of the bulbourethral glands in the male.
3. The proximal portion of the vagina forms from the fused paired Müllerian ducts; the distal portion forms from the sinovaginal bulbs, which later canalize.

4. Clitoral hypertrophy in the newborn should suggest congenital adrenal hyperplasia; other etiologies include neurofibromatosis and an androgen-producing tumor in the mother.
5. A small clitoris may be seen in androgen insensitivity syndrome.
6. In planning treatment for a transverse vaginal septum, it is of critical importance to determine whether there is a cervix and, if present, its exact location relative to the septum.
7. Vaginal atresia differs from vaginal agenesis and testicular feminization in that the Müllerian structures are not affected. As a result, the uterus, cervix, and upper portion of the vagina are normal. In vaginal agenesis the uterus is generally absent or rudimentary; there are often associated renal and skeletal abnormalities.
8. A complication of the use of skin grafts to create a neovagina is the increased incidence of vaginal stenosis and the requirement for repeated vaginal dilatation.
9. Most anomalies of lateral fusion have no functional significance.
10. When an interlabial mass appears to be associated with the urethra, workup should always include renal pelvic ultrasonography.
11. Urethral prolapse appears as a doughnut-shaped mass with the urethral orifice in the center; a mass lateral to the orifice may be a paraurethral cyst, a prolapsed ureterocele, a urethral polyp, or an ectopic ureter.
12. An imperforate hymen at birth appears as a bulge in the perineum; it may present at puberty as cyclic abdominal pain and amenorrhea.

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Disorders of Sexual Development

Etiology, Evaluation, and Medical Management

Richard Nithiphaisal Yu, and David A. Diamond

Keywords

cryptorchidism; gubernaculum; orchidopexy; subfertility; testicular descent; testicular germ cell tumor

Questions

1. Which of the following statements is TRUE about *SRY* (sex-determining region of the Y-chromosome gene)?
    a. It is synonymous with the H-Y antigen.
    b. SRY protein has a characteristic high-mobility group (HMG), DNA-binding domain.
    c. It is regulated by SOX9 expression.
    d. It was genetically mapped by the study of patients with Klinefelter and Turner syndromes.
    e. It is synonymous with Zfy in humans.
2. Which statement about steroidogenic factor 1 (SF1) is correct?
    a. The gene encodes a nuclear hormone transcription factor.
    b. It regulates the expression of anti-Müllerian hormone.

c. It regulates the expression of enzymes involved in steroid production.
d. Mutations in the gene are associated with undervirilization.
e. All of the above are true.

3. Which of the following statements is TRUE regarding anti-Müllerian-hormone (AMH)?
   a. It acts systemically to produce Müllerian regression.
   b. It is secreted by the fetal Leydig cells.
   c. It functions normally in patients with hernia uteri inguinale.
   d. It is secreted at 7 to 8 weeks of gestation, representing the initial endocrine function of the fetal testis.
   e. It is secreted by the fetal testis at 10 weeks of gestation, after testosterone production has begun.
4. Which description of *WT1* is correct?
   a. Mutations in WT1 can result in either Denys-Drash or Frasier syndrome.
   b. Mutations in WT1 are associated with adrenocortical carcinoma.
   c. Loss of WT1 function has not been associated with genitourinary anomalies.
   d. Duplication of WT1 has been associated with dosage-sensitive sex reversal.
   e. The gene was originally isolated in cloning experiments and localized to the X chromosome.
5. Which of the following statements is TRUE regarding fetal testosterone?
   a. It causes regression of the Müllerian ducts.
   b. It is produced primarily by the adrenal gland.
   c. It acts locally to virilize the urogenital sinus and genital tubercle.
   d. It acts locally to virilize the internal Wolffian duct structures.
   e. It enters target tissue by active diffusion.
6. Which of the following statements is TRUE regarding dihydrotestosterone (DHT)?
   a. It produces virilization of Wolffian duct structures.
   b. It is converted by 5α-reductase to testosterone in the liver.

c. It induces virilization of the urogenital sinus.
d. It acts locally to produce regression of Müllerian structures.
e. It is secreted in large quantities by the fetal testis.

7. Which of the following statements is TRUE regarding patients with Klinefelter syndrome?
   a. They have at least one X and two Y chromosomes.
   b. They are at increased risk for development of adenocarcinoma of the breast.
   c. They undergo replacement of Leydig cells with hyaline.
   d. They are normally fertile.
   e. They bear little resemblance to XX males.
8. The streak gonad of Turner syndrome:
   a. can descend to the scrotum.
   b. has a reduced number of oocytes.
   c. results in increased risk for development of seminoma since only one X chromosome is present.
   d. results in increased risk for development of gonadoblastoma when Y chromosome elements are present.
   e. is located in the round ligament.
9. Which of the following statements is TRUE regarding patients with 46, XY complete gonadal dysgenesis?
   a. They frequently have chromosomal anomalies.
   b. They are at significant risk for gonadal tumors.
   c. They have somatic defects similar to Turner syndrome.
   d. They have gonadal histology different from that of patients with Turner syndrome.
   e. They derive similar benefit from synthetic growth hormone as do patients with Turner syndrome.
10. What is the most common finding in ALL cases of Denys-Drash syndrome?
   a. Gonadoblastoma
   b. Nephropathy with early-onset proteinuria
   c. Wilms tumor
   d. Calyceal blunting
   e. Progressive renal failure

11. Which of the following statements is TRUE regarding patients with embryonic testicular regression or bilateral vanishing testes syndromes?
    a. They have normal testosterone and elevated estradiol levels.
    b. They have normal testosterone but decreased DHT levels.
    c. They have castrate testosterone and elevated gonadotropin levels.
    d. They have castrate testosterone and decreased gonadotropin levels.
    e. They have normal follicle-stimulating hormone but decreased luteinizing hormone levels.
12. Which of the following statements is TRUE regarding the ovotestis in ovotesticular DSD?
    a. It cannot descend from the retroperitoneum.
    b. It is found in the minority of patients.
    c. It can be unilateral or bilateral.
    d. It has testicular and ovarian elements randomly distributed.
    e. It is impossible to cleave surgically.
13. An important consideration for gender assignment in the ovotesticular DSD patient is:
    a. the potential for fertility.
    b. the impossibility of precisely dividing an ovotestis surgically.
    c. the high incidence of gonadal tumors in 46, XX ovotesticular DSD.
    d. the familial pattern of inheritance of the disorder.
    e. androgen insensitivity.
14. Which of the following statements is TRUE regarding the 21-hydroxylase deficiency in congenital adrenal hyperplasia (CAH)?
    a. It accounts for 50% of CAH cases.
    b. It occurs as a result of *CYP21A* inactivation in the majority of cases.
    c. It occurs with simple virilization in 75% of cases and salt wasting in 25% of cases.
    d. It occurs with a predictable phenotype.
    e. It is transmitted in an autosomal dominant pattern.
15. Prenatal treatment of fetuses with CAH using dexamethasone:

a. is appropriate therapy in the majority of at-risk fetuses.
b. can be initiated after a diagnosis of CAH is confirmed by amniocentesis.
c. is of no risk to the fetus.
d. has been demonstrated to be effective.
e. acts by suppressing maternal ACTH.

16. Which of the following statements is TRUE regarding enzymatic disorders of testosterone biosynthesis?
    a. They are transmitted in an autosomal dominant pattern.
    b. They are associated with persistent Müllerian structures.
    c. They appear clinically with a predictable phenotype.
    d. They may involve impaired glucocorticoid and mineralocorticoid synthesis.
    e. They may be associated with fertility.
17. Which of the following statements is TRUE regarding patients with complete androgen insensitivity?
    a. They are typically raised as female.
    b. They have normal Wolffian duct structures.
    c. They have persistent Müllerian duct structures.
    d. They should undergo orchiectomy as early as possible.
    e. They have a 2% incidence of inguinal hernia.
18. Partial androgen insensitivity syndrome is characterized as:
    a. a group of defects in testosterone biosynthesis.
    b. a form of 5α-reductase deficiency.
    c. Y-chromosome mutations.
    d. a disorder of androgen receptor quantity or function.
    e. a disorder transmitted in an autosomal recessive manner
19. Which of the following statements is TRUE regarding patients with 5α-reductase deficiency?
    a. Patients with a 46,XY karyotype are always infertile.
    b. Isoenzymes 1 and 2 are abnormal.
    c. Serum testosterone levels are normal, but there is a decreased testosterone/DHT ratio.
    d. Masculinization occurs at puberty.
    e. Prostatic enlargement occurs at puberty.

20. Which of the following statements is TRUE regarding patients with persistent Müllerian duct syndrome?
    a. They have absent Wolffian duct structures.
    b. They represent a homogeneous disorder of involving the AMH receptor.
    c. They should undergo routine removal of Müllerian structures.
    d. They experience a high incidence of transverse testicular ectopia.
    e. They are uniformly infertile.
21. Which of the following statements is TRUE regarding Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome?
    a. It presents most commonly as infertility.
    b. It is a homogeneous disorder entailing congenital absence of the uterus and vagina.
    c. It is associated with a spectrum of ovarian abnormalities.
    d. It has associated upper urinary tract anomalies, more frequently with the atypical disorder.
    e. It is associated with persistent Wolffian duct structures.
22. In a neonate with hypospadias and a unilateral cryptorchid testis:
    a. midshaft location of the urethral meatus is an important risk factor for DSD.
    b. impalpability of the cryptorchid testis carries a 50% risk of a DSD.
    c. palpability of the cryptorchid testis effectively rules out DSD.
    d. perineal hypospadias is not a risk factor for a DSD.
    e. difference in tissue texture of the poles of the cryptorchid gonad is suggestive of tumor.
23. Gender identity:
    a. is synonymous with gender role.
    b. is primarily determined by prenatal exposure to androgens.
    c. is primarily determined by postnatal environmental influences.
    d. is defined as the identification of self as either male or female.
    e. does not play a role in gender dysphoria.

Answers

1. **b. SRY protein has a characteristic high-mobility group (HMG), DNA-binding domain.** This domain can induce significant DNA bending when bound to the regulatory regions of target genes. *SRY* is not synonymous with H-Y antigen. **The *SRY* gene is located on the short arm of the Y chromosome adjacent to the pseudoautosomal boundary.** SRY regulates SOX9 expression during gonadal differentiation.
2. **e. All of the above are true.** The expressed SF1 protein is a nuclear hormone transcription factor that binds to DNA. Several studies have demonstrated that SF1 regulates the transcription of AMH and multiple steroidogenic enzyme genes. Recent studies suggest that mutations in *SF1*, also known as *NR5A1*, have an additional role in DSD and are associated with varying levels of undervirilization, adrenal insufficiency, and gonadal dysgenesis.
3. **d. It is secreted at 7 to 8 weeks of gestation, representing the initial endocrine function of the fetal testis.** AMH is produced by the Sertoli cells and acts in a paracrine fashion to induce regression of Müllerian structures. Hernia uteri inguinale, also known as persistent Müllerian duct syndrome (PMDS), occurs in patients with a 46, XY karyotype and normal male external genitalia. These patients have persistent internal Müllerian duct structures that may herniate or be associated with an undescended testicle.
4. **a. Mutations in *WT1* can result in either Denys-Drash or Frasier syndrome.** Denys-Drash syndrome is characterized by a triad of Wilms tumor, congenital nephropathy, and genital atypia or Difference of Sex Development (DSD). Frasier syndrome is associated with gonadoblastoma, congenital nephropathy, and gonadal dysgenesis. Patients with Frasier syndrome do not have a known predisposition to Wilms tumor. The *WT1* gene is located on chromosome 11 and is not associated with dosage-sensitive sex reversal.
5. **d. It acts locally to virilize the internal Wolffian duct structures.** Androgens are essential for virilization of Wolffian duct structures, the urogenital sinus, and genital tubercle. Testosterone, the major androgen secreted by the testes, enters target tissues by passive diffusion. The local production and paracrine effect of androgen is

important for Wolffian duct development, which does not occur if testosterone is supplied only via the peripheral circulation. Masculinization of the external genitalia results from systemic delivery of testosterone and local conversion to DHT.

6. **c. It induces virilization of the urogenital sinus.** In some cells, such as those in the urogenital sinus, testosterone is converted to DHT by intracellular 5α-reductase. Testosterone or DHT then binds to a high-affinity intracellular receptor protein, and this complex enters the nucleus, where it binds to acceptor sites on DNA, resulting in target gene activation and protein synthesis. **Therefore, in tissues equipped with 5α-reductase at the time of sexual differentiation, such as prostate, urogenital sinus, and external genitalia, DHT is the active androgen.**
7. **b. They are at increased risk for development of adenocarcinoma of the breast.** Gynecomastia, which can be quite marked, is a common pubertal development in patients with Klinefelter syndrome. As a result, these patients are at **eight times the risk for developing breast carcinoma** relative to normal males. Males with Klinefelter syndrome have one Y chromosome and at least two X chromosomes. The most common karyotype is 47, XXY. The cells lining the seminiferous tubules undergo replacement with hyaline during pubertal development.
8. **d. Results in increased risk for development of gonadoblastoma when Y chromosome elements are present.** In patients with occult Y chromosome material, the risk of gonadoblastoma, an *in situ* germ cell cancer, is approximately 30%. The streak gonad is usually abdominal in location, hypoplastic, and predominantly composed of fibrous tissue.
9. **b. They are at significant risk for gonadal tumors.** Up to 35% develop tumors by 30 years of age. Patients with 46,XX complete gonadal dysgenesis are closely related to those with Turner syndrome. But, they do not have somatic defects associated with Turner syndrome (e.g., broad chest, neck webbing, cardiac and renal anomalies, and short stature).
10. **b. Nephropathy with early-onset proteinuria.** The full triad of the syndrome includes nephropathy, characterized by the early

onset of proteinuria, hypertension, and progressive renal failure in the majority, Wilms tumor, and DSD. Denys-Drash is caused by mutations in the Wilms tumor suppressor (*WT1*) gene. Because incomplete forms of the syndrome may occur, the nephropathy aspect is regarded as the common denominator of the syndrome.

11. **c. They have castrate testosterone and elevated gonadotropin levels.** The diagnosis can be made on the basis of a 46,XY karyotype and castrate levels of testosterone, despite persistently elevated serum luteinizing hormone and follicle-stimulating hormone levels. This can be detected during "mini-puberty" from 2 to 6 months of age. Serum AMH is a useful marker to detect the presence of testicular tissue and is undetectable in these males.
12. **c. It can be unilateral or bilateral.** Ovotesticular DSD patients are individuals having both testicular tissue with well-developed seminiferous tubules and ovarian tissue with primordial follicles. The gonads may take the form of one ovary and one testis or, more commonly, one or two ovotestes. Histopathology of the ovotestis will typically demonstrate well-developed ovarian tissue and a dysgenetic testicular component. The ovotestis may be located at any point along the path of testicular descent. Ovotestes with bipolar distribution of tissue may often be surgically separated.
13. **a. The potential for fertility.** The most important aspect of management in ovotesticular DSD is gender assignment. An important factor to consider is the potential for fertility since individuals with ovotesticular DSD have the potential for fertility if raised as female with the appropriate ductal structures. Pregnancies have been reported in these women. Testicular tissue in an ovotestes is commonly dysgenetic. Approximately 3% of patients with 46, XY ovotesticular DSD develop gonadal tumors. Tumors are rare in patients with 46, XX ovotesticular DSD.
14. **b. It occurs as a result of *CYP21A2* inactivation in the majority of cases.** 21-Hydroxylase deficiency accounts for 95% of CAH cases, with 75% of patients presenting with salt wasting and 25% with simple virilization. Mutations leading to gene conversion of the active *CYP21A2* gene into the inactive gene occur in 65% to 90% of cases of the classic disorder (salt wasting and simple

virilizing) and all cases of nonclassic 21-hydroxylase deficiency. 21-hydroxylase deficiency is transmitted in an autosomal recessive pattern.

15. **d. Has been demonstrated to be effective.** Treatment should be initiated before 9 weeks after the last menstrual period, once pregnancy is confirmed. A number of series have established the effectiveness of prenatal treatment of CAH with dexamethasone, which suppresses fetal secretion of adrenocorticotropic hormone. **However, a diagnosis of CAH cannot be confirmed before therapy is initiated because the diagnosis is usually made by chorionic villus sampling (9 to 11 weeks gestation) or amniocentesis (16 to 17 weeks gestation). Therefore, if treatment is initiated for all at-risk fetuses, the majority (seven out of eight) may be treated unnecessarily before confirmatory diagnosis.** Noninvasive prenatal diagnosis of CAH using cell-free fetal DNA obtained from maternal plasma is a promising new development that may allow identification of male fetuses, who would not benefit from treatment, well before 9 weeks gestation.
16. **d. They may involve impaired glucocorticoid and mineralocorticoid synthesis.** A defect in any of the five enzymes required for the conversion of cholesterol to testosterone can cause incomplete (or absent) virilization of the male fetus during embryogenesis. The first three enzymes (cholesterol side chain cleavage enzyme, 3β-hydroxysteroid dehydrogenase, and 17α-hydroxylase) are present in both the adrenals and the testes. Therefore, their deficiency results in impaired synthesis of glucocorticoids and mineralocorticoids in addition to testosterone. These enzymatic disorders are transmitted in an autosomal recessive pattern and appear clinically with a variable phenotype. Since testosterone biosynthesis is impacted, high concentration of testosterone within the testes is absent thus impacting spermatogenesis and fertility.
17. **a. They are typically raised as female.** It is of great interest that, currently, all studies of patients with complete androgen insensitivity support an unequivocal female gender identity, consistent with androgen resistance of brain tissue as well. To date

there has been only one report of a patient raised as a female who needed gender reassignment to male. [a]

18. **d. A disorder of androgen receptor quantity or function.** Androgen receptor studies in cultured fibroblasts have demonstrated two forms of receptor defect in the partial androgen insensitivity syndrome. These include a reduced number of normally functioning androgen receptors and normal receptor number but decreased binding affinity. The androgen receptor gene is located on the long arm of the X chromosome, Xq12. Partial androgen resistance is, therefore, an X-linked disorder of incomplete masculinization.
19. **d. Masculinization occurs at puberty.** At puberty, partial masculinization occurs with an increase in muscle mass, development of male body habitus, increase in phallic size, and onset of erections. However, prostatic enlargement does not occur as this is dependent on DHT production. **The type 2 isoenzyme is affected in patients with 5α-reductase deficiency, resulting in an increased testosterone/DHT ratio owing to a reduced testosterone-to-DHT conversion rate.** There are reports of sperm production and fertility in some affected individuals.
20. **d. They experience a high incidence of transverse testicular ectopia.** Persistent Müllerian duct syndrome is thought to be etiologically important in transverse testicular ectopia, occurring in 30% to 50% of cases. Aberrant AMH function may be secondary to defects in the gene for AMH or for the AMH receptor. Individuals with a 46, XY karyotype develop Wolffian duct structures normally and may have impaired spermatogenesis associated with undescended testes.
21. **d. It has associated upper urinary tract anomalies, more frequently with the atypical disorder.** MRKH syndrome is a rare congenital disorder associated with absence of the uterus and vagina (typical variant). Patients commonly present with primary amenorrhea. The atypical form of MRKH syndrome (10% of cases) includes patients with uterine remnants and/or aplasia of one or

both fallopian tubes. Urinary tract anomalies occur more commonly in patients with the atypical form of the disorder.

22. **b. Impalpability of the cryptorchid testis carries a 50% risk of a DSD.** With hypospadias and either unilateral or bilateral cryptorchidism, the incidence of a DSD is 30% overall, 15% if the undescended testis is palpable, and 50% if it is impalpable.
23. **d. Is defined as the identification of self as either male or female.** Gender role refers to aspects of behavior that distinguish males and females. The development of gender identity is poorly understood, but is influenced by prenatal and postnatal factors. Individual conflicts with gender identity are central to the concept of gender dysphoria.

Chapter Review

1. *SRY* initiates testicular organogenesis. The *SRY* gene is located on the short arm of the Y chromosome.
2. The prostate, urogenital sinus, and external genitalia are all sensitive to DHT.
3. Estrogens are not required for normal female differentiation.
4. Patients with Klinefelter syndrome have eunuchoidism, gynecomastia, azoospermia, small testes, and are tall for their age. Muscle development is often poor.
5. Patients with Turner syndrome (XO) have sexual infantilism, webbed neck, and cubitus valgus; are of the female phenotype; are short in stature; and lack secondary sexual characteristics.
6. In patients with Turner syndrome, any Y-chromosome material increases the risk for the streak gonad developing a gonadoblastoma; these patients also have increased incidence of abnormalities of the kidney, including horseshoe kidney.
7. The diagnosis of the salt-wasting type of CAH is made by obtaining a serum 17-hydroxyprogesterone value 3 to 4 days after birth. If it is elevated, the patient has CAH.
8. In patients who have severe forms of CAH difficult to control medically, bilateral adrenalectomy may be the most effective

approach.

9. A distinctly palpable gonad along the pathway of descent is highly suggestive of a testis or rarely of an ovotestis.
10. Patients with bilateral impalpable testes or a unilateral impalpable testis and hypospadias should be regarded as having a disorder of sexual development until proven otherwise whether or not the genitalia appear ambiguous. They should have a karyotype.
11. For normal ovarian development there must be two X chromosomes.
12. In the developing embryo, MIS and testosterone act locally and unilaterally.
13. Mixed gonadal dysgenesis is the second most common cause of DSD after CAH, and it is characterized by a unilateral testis, a contralateral streak gonad, and persistent Müllerian structures with varying degrees of inadequate masculinization.
14. The syndrome of complete androgen insensitivity is characterized by 46,XY karyotype, bilateral testes, female-appearing external genitalia, and absence of Müllerian structures; the testes are prone to tumors, which usually occur after puberty in 1% to 2% of affected individuals.
15. Persistent Müllerian duct syndrome is due to a defect in the gene for AMH or for the AMH type II receptor in patients with 46,XY karyotype. These patients have normal male external genitalia and internal Müllerian duct structures (fallopian tubes, uterus, and upper vagina).
16. The finding of a palpable gonad in a newborn with ambiguous genitalia effectively rules out overt masculinization of the female (CAH).
17. Patients with Klinefelter syndrome are at eight times the risk for breast carcinoma relative to normal males.
18. Serum MIS is a useful marker for the presence of testicular tissue in the newborn period.
19. Histopathology of the ovotestis will typically demonstrate well-developed ovarian tissue and a dysgenetic testicular component.
20. Patients with complete androgen insensitivity support an unequivocal female gender identity. However, there are a few case reports of

patients with male gender identity. One patient required complete sex reassignment with both androgen treatment and phalloplasty.

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[a] T'Sjoen G, De Cuypere, G, Monstrey S, et al: Male gender identity in complete androgen insensitivity syndrome. *Arch Sex Behav* 40:635—638, 2011.

SECTION F

Reconstruction and Trauma

OUTLINE

49. Surgical Management of Disorders of Sexual Differentiation and Cloacal and Anorectal Malformations

50. Adolescent and Transitional Urology

51. Urologic Considerations in Pediatric Renal Transplantation

52. Pediatric Genitourinary Trauma

49

Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

Richard C. Rink

Keywords

cloaca; cloacal anomaly; disorders of sex development; DSD; urogenital sinus

Questions

1. Which of the following statements is FALSE regarding the construction of a neovagina utilizing bowel?
    a. Failure to develop an adequate space between the rectum and bladder can result in compromised blood flow to the segment used for vaginal construction.
    b. In general, colon is preferred versus ileum because of its lower incidence of associated postoperative stenosis.
    c. When compared with the McIndoe procedure, the bowel vagina suffers from a higher incidence of postoperative stenosis.
    d. An advantage of a bowel vagina versus the McIndoe procedure includes the lubricating properties of mucus (which

may help facilitate intercourse).
   e. One specific indication for the use of ileum is a previous history of pelvic radiation.

2. Urogenital sinus anomalies in differences of sexual development states are most commonly seen in association with:
   a. congenital adrenal hyperplasia.
   b. mixed gonadal dysgenesis.
   c. ovotesticular disorder.
   d. cloacal anomalies.
   e. gonadal dysgenesis.
3. The most common finding in cloacal anomalies that have been diagnosed by antenatal ultrasonography is:
   a. ascites.
   b. distended rectum.
   c. distended bladder.
   d. distended vagina.
   e. distended bladder and rectum.
4. What is the most common vaginal anatomy in cloacal malformation?
   a. Single vagina, single uterus
   b. Single vagina, double uterus
   c. Two vaginas, two uteri
   d. Two vaginas, one uterus
   e. Single vagina, no uterus
5. Neonatal vaginoplasty combined with clitoroplasty and labiaplasty has all of the following advantages EXCEPT:
   a. it allows phallic skin for vaginal reconstruction.
   b. maternal estrogens increase vaginal thickness and vascularity.
   c. tissues are less scarred.
   d. vaginal stenosis is clearly less.
   e. clitoroplasty must retain normal clitoral innervation while providing excellent cosmesis.
6. The cut-back vaginoplasty is appropriate for:
   a. labial fusion.
   b. low vaginal confluence.
   c. high vaginal confluence.

d. vaginal atresia.
e. vaginal agenesis.

7. Surgical management of cloacal malformations involves all of the following steps EXCEPT:
   a. decompression of the gastrointestinal tract.
   b. decompression of the genitourinary tract.
   c. vaginostomy.
   d. definitive repair of the cloaca.
   e. correction of nephron destructive anomalies.
8. Fecal continence after cloacal reconstruction is most closely related to:
   a. the level of rectal confluence.
   b. associated urinary anomalies.
   c. neurologic status.
   d. the type of repair.
   e. the timing of the repair.

Answers

1. **c. When compared with the McIndoe procedure, the bowel vagina suffers from a higher incidence of postoperative stenosis.** A high incidence of postoperative vaginal stenosis necessitates postoperative vaginal dilatation in the McIndoe procedure.
2. **a. Congenital adrenal hyperplasia.** Urogenital sinus abnormalities are most often seen in disorders of sexual differentiation states, most commonly in association with congenital adrenal hyperplasia, which has been noted to have an incidence as frequent as 1 in 500 in the nonclassic mild forms.
3. **d. Distended vagina.** The common finding in all reports has been a cystic pelvic mass between the bladder and rectum, representing a distended vagina.
4. **c. Two vaginas, two uteri.** In Hendren's report on 154 patients with cloacal anomalies, 66 patients had one vagina, 68 had two vaginas, and the vagina was absent in 20. The incidence of vaginal duplication is even higher in the author's own patient population.

The uterus anomaly generally is similar to the vaginal anomaly, that is, two vaginas with two uteri.

5. **d. Vaginal stenosis is clearly less.** Other investigators, including the author's group, have thought that vaginoplasty, regardless of the vaginal location, is best combined with clitoroplasty in a single stage. This allows the redundant phallic skin to be used in the reconstruction, adding flexibility for the surgeon, which is compromised when the skin has been previously mobilized. Furthermore, the authors and others have noted that maternal estrogen stimulation of the child's genitalia results in thicker vaginal tissue, which is better vascularized, making vaginal mobilization more easily performed.
6. **a. Labial fusion.** The cut-back vaginoplasty is rarely used and is appropriate only for simple labial fusion.
7. **c. Vaginostomy.** Surgical management now involves four basic steps: decompression of the gastrointestinal tract, decompression of the genitourinary tract, correction of nephron-destructive or potentially lethal urinary anomalies, and definitive repair of the cloaca.
8. **c. Neurologic status.** Fecal continence is directly related to neurologic status.

Chapter Review

1. The communication of the vagina with the urinary tract usually occurs in the mid to distal urethra.
2. In patients with congenital adrenal hyperplasia, the location of the confluence of the vagina and the urethra is the critical determinant in the surgical management.
3. Hydrometrocolpos is frequently the initial sign of a urogenital sinus abnormality. It is caused by urine draining into the vagina with poor vaginal drainage.
4. Persistent clitoral hypertrophy may occur in premature infants without DSD.

5. A cervical impression in the dome of the vagina seen on genitography denotes normal female internal organs.
6. When gonads in the neonatal period require biopsy, a deep biopsy is appropriate because the ovarian component of an ovotestis may cover the testicular component.
7. Renal anomalies commonly occur in patients with a persistent cloaca.
8. Women with CAH are less satisfied as adults with their genitalia than are controls.
9. Genital reconstruction must address clitoroplasty, labiaplasty, and vaginoplasty.
10. When clitoroplasty is performed, the glans, tunics, and neurovascular bundles should be preserved. The neurovascular bundles should not be mobilized.
11. The timing of surgery for genital reconstruction is controversial: vaginoplasty is best combined with clitoroplasty and labiaplasty as a single procedure.
12. Vaginoplasty may be performed with a posterior-based perineal flap for low vaginal confluence, a pull-through vaginoplasty for a high confluence, and vaginal replacement for an absent or rudimentary vagina.
13. The flap in a "flap vaginoplasty" must reach the normal caliber of the vagina; that is, it must be placed cephalad to the narrowed area of the distal vagina.
14. A vaginoplasty performed in the neonatal period will usually require a secondary procedure after puberty.
15. In cloacal malformations, surgical management initially involves decompression of the gastrointestinal tract, generally with a colostomy, and decompression of the genitourinary tract. This is followed by correction of urinary collecting system abnormalities that impair urine flow. A single-stage repair of rectal, vaginal, and urethral abnormalities is performed at a later date when the child is stable.
16. Intermittent catheterization of the vagina may successfully decompress the genitourinary tract in cloacal malformations.
17. Spinal cord abnormalities are frequently found in patients with persistent cloaca.

18. In patients who have had corrective surgery for cloacal abnormalities, a high percentage have a neuropathic component to both urinary and fecal incontinence.
19. In planning treatment for a transverse vaginal septum, it is critical to determine whether there is a cervix and, if present, its exact location relative to the septum.
20. Vaginal atresia differs from vaginal agenesis and testicular feminization in that the Müllerian structures are not affected. As a result, the uterus, cervix, and upper portion of the vagina are normal.

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Adolescent and Transitional Urology

Dan Wood

Keywords

adolescence; adolescent urology; transitional urology

Questions

1. Patients with complex congenital urologic conditions:
    a. are unlikely to require surgery in adulthood.
    b. prefer to be looked after in a general urology clinic.
    c. generally do not work as adults.
    d. desire to be normal.
    e. can usually be discharged from care after 18 years of age.
2. Effective transition requires:
    a. a clear plan by the age of 14 years.
    b. a purposeful process to get a young adult to take responsibility for their health.
    c. preparation of the family.
    d. trust between pediatric and adolescent/adult teams.
    e. all of the above.
3. Adolescent urology training requires:
    a. a fellowship in plastic surgery.
    b. board certification (or equivalent) in urology.
    c. a fellowship in adolescent medicine.

d. 1 year of training in psychology.
e. all of the above.

4. Regarding young adults with congenital anomalies of the urogenital system:
    a. boys with posterior urethral valves commonly have a dorsal chordee.
    b. all young men with corrected hypospadias need lifelong follow-up.
    c. an augmentation cystoplasty can lead to a false-positive urinary pregnancy test.
    d. patients with spina bifida often die as a result of renal failure.
    e. in a boy with exstrophy, the testes are usually badly damaged.
5. In which of the following situations should elective cesarean section be performed for urologic indications?
    a. With an artificial urinary sphincter
    b. Women with exstrophy and an augmentation cystoplasty
    c. Women with a glomerular filtration rate (GFR) below 40 $mL/min/m^2$
    d. Women with a neuropathic bladder
    e. Women with simple (nonsalt-wasting) congenital adrenal hyperplasia
6. In boys with posterior urethral valves:
    a. intervention in utero will prevent the need for renal transplant in adulthood.
    b. fertility will likely be normal.
    c. 60% will have urinary incontinence.
    d. there is no increased risk of proteinuria.
    e. bladder function will be normal if valves are incised in the first month of life.

Answers

1. **d. Desire to be normal.** This a fundamental element of looking after patients with major congenital urologic anomalies. As they move into adulthood, they commonly express a desire to be normal

and may exert their independence to try and appear more so—i.e., by not coming to a hospital or not catheterizing. Many will need lifelong care, many work, many prefer to be looked after in a clinic with an expertise in their (rare) condition, and a significant number will need surgery as adults.

2. **e. All of the above.** Planning and preparation definitely make life easier for patients, parents, and those looking after the patients as they move into adulthood. It appears to significantly reduce anxiety and improve engagement, and trust between the team who have looked after the patient in childhood and the adult team is important.
3. **b. Board certification (or equivalent) in urology.** This is the only requirement at present. A urology training program will offer most of the skills required. Important elements are training in complex pediatric urology and reconstructive urology. Psychology can be very important, hence it is important to have a trained member on the multidisciplinary team.
4. **c. An augmentation cystoplasty can lead to a false-positive urinary pregnancy test.** Data suggest this happens in 57% of augmentation cystoplasty cases. The testes are normal in exstrophy, valve boys would be expected to have a normal penis, mild forms of hypospadias do not usually need long-term follow-up—just information on who to contact if they do have problems. With aggressive early bladder management (clean intermittent catheterization [CIC] and antimuscarinics) deaths from renal failure have dramatically decreased in patients affected by spina bifida.
5. **b. Women with exstrophy and an augmentation cystoplasty.** A cystoplasty on its own is not an absolute indication for a caesarean section—if the pelvis is otherwise normal, a vaginal delivery may be possible (this is an obstetric decision). Fundamentally, it is important to avoid a caesarean section without appropriate urologic support—elective sections should be planned with such a urologist present, and unless such expertise is available 24 hours a day seven days a week, then an emergency caesarean section should be avoided.

6. **b. Fertility will likely be normal.** For many years this was not thought to be the case. More recent data suggest that fertility is likely to be normal.

Chapter Review

1. Transition is the development of a young adult's understanding of and independence in their healthcare.
2. Preparation for transition can occur over a number of years.
3. A multidisciplinary team is essential in both pediatric and adult care.
4. There are a variety of models for delivering transition and adolescent care—these will vary depending on the healthcare system.
5. Those caring for adult patients with complex congenital anomalies must be trained in both pediatric and adult reconstructive urology.
6. Congenital lifelong urology requires a proactive approach and an understanding of complex congenital genitourinary conditions.
7. Regular follow up examining renal and urinary function are essential for many.
8. Sexual function and fertility must be proactively addressed and managed.
9. Pregnancy is likely to need preparation (for some) and combined urologic and obstetric care.
10. Collaboration with other specialties is essential for safe and appropriate care.

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Urologic Considerations in Pediatric Renal Transplantation

Craig A. Peters, and Armando J. Lorenzo

Keywords

bladder; complications; kidney transplant; urinary diversion

Questions

1. An indication for urodynamic evaluation of a child being prepared for renal transplant is:
    a. poststreptococcal glomerulonephritis-induced renal failure.
    b. ongoing grade III vesicoureteral reflux.
    c. end-stage renal disease (ESRD) in a child with nocturnal enuresis.
    d. prolonged anuria after focal segmental glomerulosclerosis (FSGS) renal failure.
    e. ESRD in a child with wetting but no urinary tract infection (UTIs).
2. Renal transplantation into a child with an augmented bladder is associated with:
    a. cyclosporine toxicity.
    b. hyperchloremic metabolic alkalosis.
    c. refractory rejection.
    d. recurrent infection and risk of sepsis.
    e. no significant change in graft survival.

3. In a 10-year-old boy with a 50-mL capacity bladder, a history of posterior urethral valves, ESRD, and anuria, optimal initial bladder management in preparation for renal transplantation is:
    a. ileocystoplasty at time of renal transplantation.
    b. gastrocystoplasty and continent catheterizable stoma.
    c. bladder cycling by intermittent catheterization.
    d. antimuscarinic medications and proceeding with transplantation.
    e. transplant into transverse colon loop and delayed augmentation.
4. Indications for pre-transplant native nephrectomy include all of the following EXCEPT:
    a. refractory two-drug hypertension.
    b. multicystic dysplastic kidney (MCDK).
    c. persisting grade III VUR in a 4-year-old.
    d. age under 12 months.
    e. ESRD associated with Denys-Drash syndrome.
5. A nonrefluxing transplant ureteroneocystostomy should be performed:
    a. in infants only.
    b. in any child undergoing renal transplantation.
    c. only in children on intermittent catheterization.
    d. in children at risk for UTI or bladder dysfunction.
    e. only in children with neurogenic bladder dysfunction.
6. Following a cadaveric renal transplant for FSGS, a 10-year-old boy is found to have distal ureteral obstruction from a 4-cm stricture that fails balloon dilation and stenting. The best option for management is:
    a. ileal interposition.
    b. Boari flap to graft pelvis.
    c. transplant nephrectomy.
    d. psoas hitch transplant ureteroneocystostomy.
    e. transplant ureter to native ureteroureterostomy.
7. Six months following an uncomplicated living related donor renal transplant for FSGS and prior native nephrectomies, a 7-year-old girl develops acute pyelonephritis of the graft with a rise in

creatinine. There is no hydronephrosis on ultrasound. A voiding cystourethrogram (VCUG) demonstrates grade III reflux into the graft, but none in the native ureters. She has normal voiding patterns. The best option for management is:

a. urodynamic evaluation and MR of the spine.
b. transplant ureter to native ureteroureterostomy.
c. endoscopic injection of subureteral bulking agent.
d. 6 months of antibiotic prophylaxis and observation.
e. continuous antibiotic prophylaxis and repeat studies in 18 months anticipating spontaneous resolution.

8. Eighteen months following an uncomplicated living related renal transplant, a 5-year-old boy with a history of posterior urethral valves develops a rising serum creatinine level, graft hydronephrosis, and two febrile UTIs. VCUG shows grade III reflux into the graft and moderate bladder trabeculation similar to his pre-transplant pattern. The next step in his care is:

a. psoas hitch ureteral reimplantation.
b. nontunneled ureteroneocystostomy.
c. ileo-cystoplasty and appendicovesicostomy.
d. transplant ureter to native ureteroureterostomy (U-U).
e. urodynamic evaluation and likely antimuscarinics and intermittent catheterization.

9. Twenty-four hours after removal of a ureteral stent, an 8-year-old boy has minimal urine output and develops right lower quadrant tenderness. Two months prior, he had a living related renal transplant following bilateral nephrectomies, radiation, chemotherapy, and surveillance for stage 5 Wilms tumors. A Foley catheter was placed, draining 5 mL of clear urine in 2 hours. He underwent urgent ultrasound evaluation (Fig. 51.1). What is the next step in management?

a. Replacement of ureteral stent
b. Renal (allograft) biopsy
c. Percutaneous nephrostomy placement
d. Percutaneous drainage of fluid collection
e. Treat for rejection with high-dose steroids

Answers

1. **e. ESRD in a child with wetting but no urinary tract infections (UTIs). The child with ongoing wetting is the most likely to have treatable voiding dysfunction that may put the renal graft at risk.** Simple bladder defunctionalization with no history of underlying bladder dysfunction is unlikely to need urodynamic evaluation and often may not need bladder cycling. Reflux alone without recurrent UTI, wetting, or a neurogenic cause is not likely to benefit from urodynamics. Nocturnal enuresis is not an indication for invasive evaluation of bladder function.



FIG. 51.1

2. **e. No significant change in graft survival. Although early reports implied a high risk due to bladder augmentation in children undergoing renal transplant, modern series have demonstrated the safety, and indeed the benefits, of providing a low-pressure urinary reservoir on graft function.** The incidence

of positive urine cultures can be increased, but clinically significant infections are not markedly increased.

3. **c. Bladder cycling by intermittent catheterization. In the child with anuria, bladder function cannot be easily assessed until the bladder has been cycled.** While this child may require augmentation, that cannot be determined until after a trial of cycling is attempted. Transplanting into a diversion is not an acceptable alternative in a child who has the potential for adequate bladder function with medical management and intermittent catheterization.
4. **b. Multicystic dysplastic kidney (MCDK). All of these clinical situations can justify native nephrectomy except for the presence of a MCDK.** There is minimal to no risk of leaving the dysplastic kidney in place. Persisting reflux may be a relative indication, and some practitioners may choose to reimplant the native ureter or simply leave it alone if there is no significant history of infection. Denys-Drash kidneys are at increased risk for Wilms tumor formation. Children under 12 months in some centers do not routinely have native nephrectomy, although there is a slightly higher risk of graft loss without this.
5. **d. In children at risk for UTI or bladder dysfunction. Any child with a risk for UTI or who has bladder dysfunction and may therefore develop a UTI, particularly if immunosuppressed, should have a nonrefluxing ureteroneocystostomy.** Some centers perform a nonrefluxing ureteral implantation in all children, although this is probably not essential. Older children, those with no history of bladder dysfunction, recurrent infections, or a structural abnormality, can be effectively transplanted with a refluxing ureteroneocystostomy.
6. **d. Psoas hitch transplant ureteroneocystostomy. Several options are available for salvage of the graft in this situation, but the most effective is a psoas hitch when there is still some graft ureter available.** A transplant to native ureteroureterostomy is effective but may require nephrectomy in some cases. Boari flaps may be useful for cases in which the entire ureter is lost or if there is no native ureter remaining. Even with loss of the entire ureter, a

renal pelvis to native ureter anastomosis is effective. Ileal interposition and graft nephrectomy are not viable options.

7. **b. Transplant ureter to native ureteroureterostomy. The best option in this situation is redo open reimplantation to create an effective antireflux tunnel or transplant to native ureteroureterostomy if there is sufficient native ureter.** In the absence of a prior urologic history, urodynamic evaluation and investigation for an occult spinal dysraphism is unlikely to be of value. Spontaneous resolution is highly unlikely, and simply waiting for another episode of pyelonephritis risks graft injury. Endoscopic injection has a limited success rate in published series and the durability is undefined. In the setting where another episode of pyelonephritis is associated with significant risk to the graft, such an approach does not seem prudent.
8. **e. Urodynamic evaluation and likely antimuscarinics and intermittent catheterization. This boy is best served by a formal assessment of his bladder function with likely use of antimuscarinics to improve capacity and compliance and intermittent catheterization to provide for emptying.** One can start this therapy empirically, but having a baseline permits assessment of the treatment effect. Moving directly to augmentation without knowing if medical management can be effective is not appropriate. Simply reimplanting the ureter or performing a U-U without treating the likely bladder dysfunction is also not appropriate.
9. **d. Percutaneous drainage of fluid collection. This boy has a large fluid collection and associated hydronephrosis. The most likely cause of his oligo-anuria is extrinsic ureteral compression due to a lymphocele.** Other causes need to be considered, including urinoma formation (and active perinephric urine leak). The next step is to drain the fluid collection and send the fluid for analysis and culture. This will determine the best course of action. Stent replacement and percutaneous nephrostomy placement will help with urine drainage yet fail to secure a diagnosis. A renal biopsy is unlikely to add to the diagnosis, except in the unlikely case that this represents rejection. Similarly, high-dose steroids will

not address the most likely etiology, and can be harmful in less likely but important complications in the differential, most notably a perinephric abscess.

Chapter Review

1. Goals of urologic evaluation and management are (1) normal urinary drainage from the kidney into a reservoir; (2) a urinary reservoir that permits low-pressure storage for a socially acceptable time; (3) volitional emptying of the reservoir, with continence; (4) absence of infection; and (5) the fewest surgical procedures and patient trauma.
2. A large fraction of children in need of renal replacement will have some type of uropathy, either congenital obstruction, vesicoureteral reflux, or neuropathic bladder dysfunction.
3. Pre-transplant assessment is directed by the underlying condition and the status of the bladder and kidneys.
4. Urodynamic testing aims to assess bladder capacity, compliance, and emptying, as well as sphincter function.
5. Indications for urodynamics include a known neuropathic bladder, history of posterior urethral valves, and children with ongoing voiding dysfunction, high-grade hydronephrosis, or recurrent UTIs.
6. The most common bladder abnormality associated with uropathy and ESRD is a low-capacity, hypertonic bladder with poor compliance.
7. There is no evidence that bladder augmentation increases the risk of transplant.
8. Recurrent pyelonephritis is a potential hazard for the transplant and is associated with graft loss.
9. Initiating clean intermittent catheterization (CIC) in preparation for transplantation serves an assessment purpose as well as facilitating bladder emptying.
10. Bladder re-functionalization is often best accomplished by bladder cycling to increase capacity, determine bladder wall compliance, and assess the family's ability to perform CIC.
11. The ability to empty spontaneously will also impact the decision regarding the need for a continent catheterizable stoma.

12. Indications for augmentation prior to transplantation include capacity less than 75% of expected for age, with pressures above 30 cm H_2O, on catheterization every 3 h, and maximal antimuscarinic medications.
13. For patients on peritoneal dialysis, intraperitoneal surgery will likely require temporary transition to hemodialysis.
14. In general, any major urologic reconstruction should be undertaken well before anticipated transplantation.
15. Native nephrectomy is indicated for malignant hypertension, profound nephrotic syndrome with malnutrition due to protein loss, high risk for malignancy, recurrent upper tract infections, and massive reflux.
16. Extravesical ureteroneocystostomy is the preferred ureteral anastomosis.
17. Transplant to native ureteroureterostomy is an effective option.
18. There are no data to support routine ureteral stenting at the time of transplantation.
19. Common urologic complications for pediatric renal transplant include ureteral stenosis (6%), urinary leaks (3%), stones (2%), and clinically significant reflux (2%).
20. Urine leaks are typically identified in the early postoperative period with increasing fluid from the wound drains.
21. Urinary infection is a long-term and often delayed complication that largely reflects the status of bladder function.
22. Vesicoureteral reflux into the transplant is entirely distinct from routine reflux as the risk to renal function of acute pyelonephritis is greater.
23. In the setting of a rising creatinine level and hydronephrosis, obstruction and rejection may both be present.
24. The etiology of ureteral obstruction is usually in the distal ureter with stenosis at the reimplantation site.
25. Bladder dysfunction may produce infection, but may also create an obstructive process that impairs renal graft function.
26. Nephrolithiasis is uncommon, but may occur in up to 5% of patients.

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Pediatric Genitourinary Trauma

Bruce J. Schlomer, and Micah A. Jacobs

Keywords

bladder trauma; genitourinary trauma; kidney trauma; pediatric; renal trauma; urethral trauma; urogenital trauma

Questions

1. Which of the following is not an indication for renal trauma imaging in children?
    a. Significant deceleration injury
    b. Gunshot wound to upper abdomen
    c. Microscopic hematuria
    d. Stab wound to left flank
    e. Blunt trauma with rib fracture and flank bruising
2. Which of the following is an absolute indication for operative intervention for renal trauma?
    a. Grade 4 renal laceration
    b. Hypotension that initially responds to fluid resuscitation
    c. Renal artery occlusion from blunt trauma
    d. Pulsating hematoma at time of exploration for gunshot wound to abdomen
    e. Continued gross hematuria after superselective embolization
3. Which of the following is the preferred initial option for managing urine extravasation from a renal laceration?

a. Observation
b. Ureteral stent
c. Percutaneous drain
d. Nephrostomy tube
e. Open exploration and repair

4. What is the appropriate management for a patient with renal arterial avulsion, hemodynamic instability, and a normal contralateral kidney?
    a. Vascular surgery consult
    b. Angiography with stenting
    c. Observation
    d. Nephrectomy
    e. Repair of renal artery injury
5. What is the appropriate management for a stable patient with isolated ureteropelvic junction (UPJ) disruption?
    a. Attempt at ureteral stent
    b. Percutaneous nephrostomy tube
    c. Nephrectomy
    d. Dismembered pyeloplasty
    e. Percutaneous drain into urinoma
6. Which of the following is not a reported high-risk criterion for undergoing intervention in renal trauma?
    a. Medial extravasation of contrast on delayed CT phase
    b. Lack of contrast in ipsilateral ureter on delayed CT phase
    c. Greater than 10% of kidney parenchyma devitalized
    d. Hematoma >2.2 cm in children
    e. Extravasation of contrast during arterial phase
7. Which of the following is not part of optimal open surgical management for a renal injury?
    a. Attempt at renal salvage
    b. Closure of the collecting system with permanent sutures
    c. Control of main renal vessels
    d. Debridement of nonviable parenchyma
    e. Closure of parenchymal defect
8. Should a patient who lost a kidney due to renal injury be allowed to participate in sports?

a. Yes, but only noncontact sports
b. Yes, no restrictions
c. Yes, but use of protective equipment is recommended for contact sports
d. Yes, but need a signed waiver releasing the physician of liability
e. No sports activities should be allowed

9. Which of the following is not typically part of a nonoperative management plan for a grade 3 renal injury?
   a. Bed rest until gross hematuria resolves
   b. Routine monitoring of vital signs and blood counts
   c. Routine physical exams
   d. Routine repeat CT scan at 48 hours if stable
   e. Angiography with superselective embolization for continued transfusion requirement from renal bleeding
10. Which of the following is the best option for a complete upper ureteral laceration?
   a. Transureteroureterostomy
   b. Ureteral reimplantation with Boari flap
   c. Ureterocalicostomy
   d. Ureteroureterostomy
   e. Ureteral stent placement
11. In what type of patient should endourologic management not be attempted as a primary treatment?
   a. Ureteral contusion
   b. Partial ureteral laceration recognized intraoperatively
   c. Complete ureteral laceration recognized 10 days after injury
   d. Partial ureteral laceration recognized 5 days after injury
   e. High-velocity gunshot wound to abdomen with injuries near ureter but no apparent ureteral injury
12. Which of the following is not an indication to evaluate for bladder injury?
   a. Gross hematuria with pelvic fracture
   b. Microscopic hematuria with pelvic fracture
   c. Pelvic fracture and inability to void

d. Abdominal trauma to lower abdomen with low urine output and ascites
e. Microscopic hematuria after penetrating injury with knife over pubic bone

13. Which of the following is not an indication for open repair of bladder rupture?
    a. Extraperitoneal bladder rupture with gross hematuria
    b. Extraperitoneal bladder rupture undergoing exploratory laparotomy for unrelated injuries
    c. Extraperitoneal bladder rupture with bladder neck injury
    d. Stable patient with intraperitoneal bladder rupture and normal serum creatinine
    e. Extraperitoneal bladder rupture with vaginal laceration
14. Which of the following is not an acceptable option for initial management of urethral disruption?
    a. Gentle attempt at blind catheter placement in suspected partial urethral disruption by retrograde urethrogram
    b. Suprapubic tube placement in complete urethral disruption
    c. Cystoscopy per urethra in emergency room (ER) for catheter placement in complete urethral disruption
    d. Dual cystoscopy in operating room (OR) retrograde from urethra and antegrade from bladder in complete urethral disruption
    e. Suprapubic tube placement in partial urethral disruption after failure to place urethra catheter
15. Which of the following is FALSE regarding management of testis injuries?
    a. Testis injuries are more common in post pubertal patients.
    b. One sign of testis rupture is irregular contour of the tunica albuginea on scrotal ultrasound.
    c. Most experts agree that a testis rupture should be repaired surgically.
    d. Hematocele on scrotal ultrasound is associated with testis rupture.
    e. Homogeneous parenchyma on testis ultrasound is a sign of testis rupture.

Answers

1. **c. Microscopic hematuria. Microscopic hematuria alone is not an indication for renal trauma imaging.** Mechanism of injury (a, b, d, e) is an important indication for renal trauma imaging.
2. **d. Pulsating hematoma at time of exploration for gunshot wound to abdomen. A pulsating hematoma at time of exploration implies life-threatening ongoing renal bleedings.** This is an absolute indication for renal exploration. All of the other options are not absolute indications for operative intervention.
3. **a. Observation. Many cases of urinary extravasation will resolve with only observation.** The other options are reserved for causes that do not resolve or develop complications such as fever, ileus, flank pain, expanding urinoma, or infection.
4. **d. Nephrectomy. An emergent lifesaving operation (nephrectomy) is needed in this scenario.** There is no time for vascular surgery consultation, and repair of renal artery injury will likely not be possible and is not indicated in the setting of a normal contralateral kidney.
5. **d. Dismembered pyeloplasty. In a stable patient with UPJ disruption the preferred treatment is dismembered pyeloplasty.** The majority of these patients, however, are not stable due to associated injuries, and other treatments such as nephrostomy tube and percutaneous drains are appropriate with delayed pyeloplasty or nephrectomy.
6. **c. Greater than 10% of kidney parenchyma devitalized. High-risk criteria include a, b, d, and e.** Another high-risk criteria for undergoing intervention in renal trauma is >25% of kidney parenchyma devitalized, not >10%.
7. **b. Closure of the collecting system with permanent sutures. Absorbable sutures should be used for any collecting system repairs or closures to prevent urolithiasis formation.**
8. **c. Yes, but use of protective equipment is recommended for contact sports. Protective equipment is recommended for**

contact sports in children with solitary kidneys, according to American Academy of Pediatrics.

9. **d. Routine repeat CT scan at 48 hours if stable. A repeat CT scan at 48 hours is recommended for higher-grade renal injuries according to AUA guidelines but is controversial for high-grade injuries if they are clinically doing well as it will unlikely change management.** All of the other options are typically part of all nonoperative management plans.
10. **d. Ureteroureterostomy. For an upper ureteral laceration, typically a ureteroureterostomy is the preferred surgical treatment.** There is generally enough ureter that can be freed up for an anastomosis, and the injury is far from the bladder, making a Boari flap more difficult.
11. **b. Partial ureteral laceration recognized intraoperatively. A ureteral laceration, partial or complete, recognized intraoperatively should be repaired.**
12. **b. Microscopic hematuria with pelvic fracture. Pelvic fracture with only microscopic hematuria is not an indication for cystogram.**
13. **a. Extraperitoneal bladder rupture with gross hematuria. An uncomplicated extraperitoneal bladder rupture with gross hematuria can often be managed with indwelling Foley and observation.** If the gross hematuria cannot be managed with Foley due to clots, then open repair would be indicated.
14. **c. Cystoscopy per urethra in emergency room (ER) for catheter placement in complete urethral disruption. If an attempt is going to be made at endoscopic realignment in complete urethral disruption, this should be performed in the operating room so retrograde and antegrade endoscopy can be performed.**
15. **e. Homogeneous parenchyma on testis ultrasound is a sign of testis rupture. Heterogeneous parenchyma on testis ultrasound is a sign of testis rupture, not homogeneous parenchyma.**

Chapter Review

1. The majority of renal trauma can be managed nonoperatively. Grade 4–5 injuries can be managed nonoperatively but have higher risk of undergoing intervention.
2. Around 25% of patients with grade 3–5 injuries being managed nonoperatively will have persistent or delayed bleeding. Superselective angioembolization of bleeding vessels is very successful and is the preferred method for management. Repeat angioembolization can also be successful if initial attempt fails.
3. Around 15% of patients with a urine leak being managed nonoperatively will have indication for intervention such as ureteral stent placement and/or percutaneous drain placement.
4. Repeat CT imaging 48–72 h after injury is not needed for patients with grade 1–3 injuries who are recovering without complication. Repeat CT imaging in 48–72 h may be obtained in patients with grade 4–5 injuries but is of unclear utility if the patient is stable and recovering without apparent complication.
5. Indications for operative intervention include hemodynamic instability due to renal bleeding, pulsatile or expanding retroperitoneal hematoma at time of laparotomy without prior renal imaging, and failure of angioembolization to control renal bleeding.
6. Control of renal vascular should be performed immediately after opening Gerota fascia or prior to opening Gerota fascia.
7. Operative intervention should include attempts to salvage kidney if possible.
8. UPJ disruptions should be repaired in acute phase if patient is stable; otherwise drainage with nephrostomy tube and possible percutaneous perirenal drain with delayed repair should be performed.
9. Renal pelvis ruptures may require operative intervention but have been managed successfully with endoscopic and percutaneous urinary drainage.
10. The risk of renal scarring is negligible in grade 1–2 injuries, approximately 50% in grade 3 injuries, and 100% in grade 4–5 injuries.
11. Follow-up imaging for grade 3–5 injuries should be considered. Starting with a renal ultrasound and obtaining CT, MRI, or DMSA

scan if necessary is a common approach.

12. Hypertension is a rare occurrence after renal trauma in children and typically is only seen after grade 3–5 injuries.
13. The development of chronic kidney disease or end-stage renal disease is very rare after renal trauma even when nephrectomy is performed.
14. Patients with a normal solitary kidney can participate in sports, but protective equipment is recommended for contact sports. High-risk activities include dirt bike riding, cycling, and all-terrain vehicle riding.
15. Ureteral injuries are rare with any type of trauma but are most commonly seen with penetrating trauma.
16. A contrast-enhanced CT scan with delayed phase or a retrograde pyelogram is the preferred methods to diagnose a ureteral injury.
17. Ureteral injuries are often associated with other organ injuries in trauma cases and can be missed at initial laparotomy for trauma.
18. If a ureteral injury is diagnosed <1 week from injury, immediate repair can be performed and may be preferable if patient is stable.
19. If a ureteral injury is diagnosed >1 week from injury, an attempt at ureteral stent placement can be made, but often a nephrostomy tube with delayed open repair is needed.
20. Ureteral injuries in children are repaired similar to adults.
21. A history of pelvic trauma with gross hematuria or inability to void, and pelvic fracture necessitate urologic evaluation with retrograde urethrogram and/or cystogram.
22. Indications to repair an extraperitoneal bladder rupture include need for placement of orthopedic hardware in the field, bony pelvis fragments protruding into the bladder, involvement of the bladder neck in the injury, and concurrent abdominal surgery.
23. Intraperitoneal bladder ruptures are managed with surgical repair.
24. Bladder neck injuries are more common in children and best served with early repair.
25. Genital injury in children should raise suspicion for abuse, and if abuse is suspected, concomitant injury to the rectum and anus should be ruled out.

26. Most circumcision injuries can probably be avoided by ensuring that adhesions are properly reduced at the beginning of the procedure.
27. Findings on ultrasound consistent with testicular rupture should prompt surgical exploration.
28. Straddle injuries make up the majority of vaginal injuries. Injuries to bladder, urethra, anus, rectum, hymen, and internal vaginal structures are suggestive of penetrative injury and should prompt evaluation for abuse.